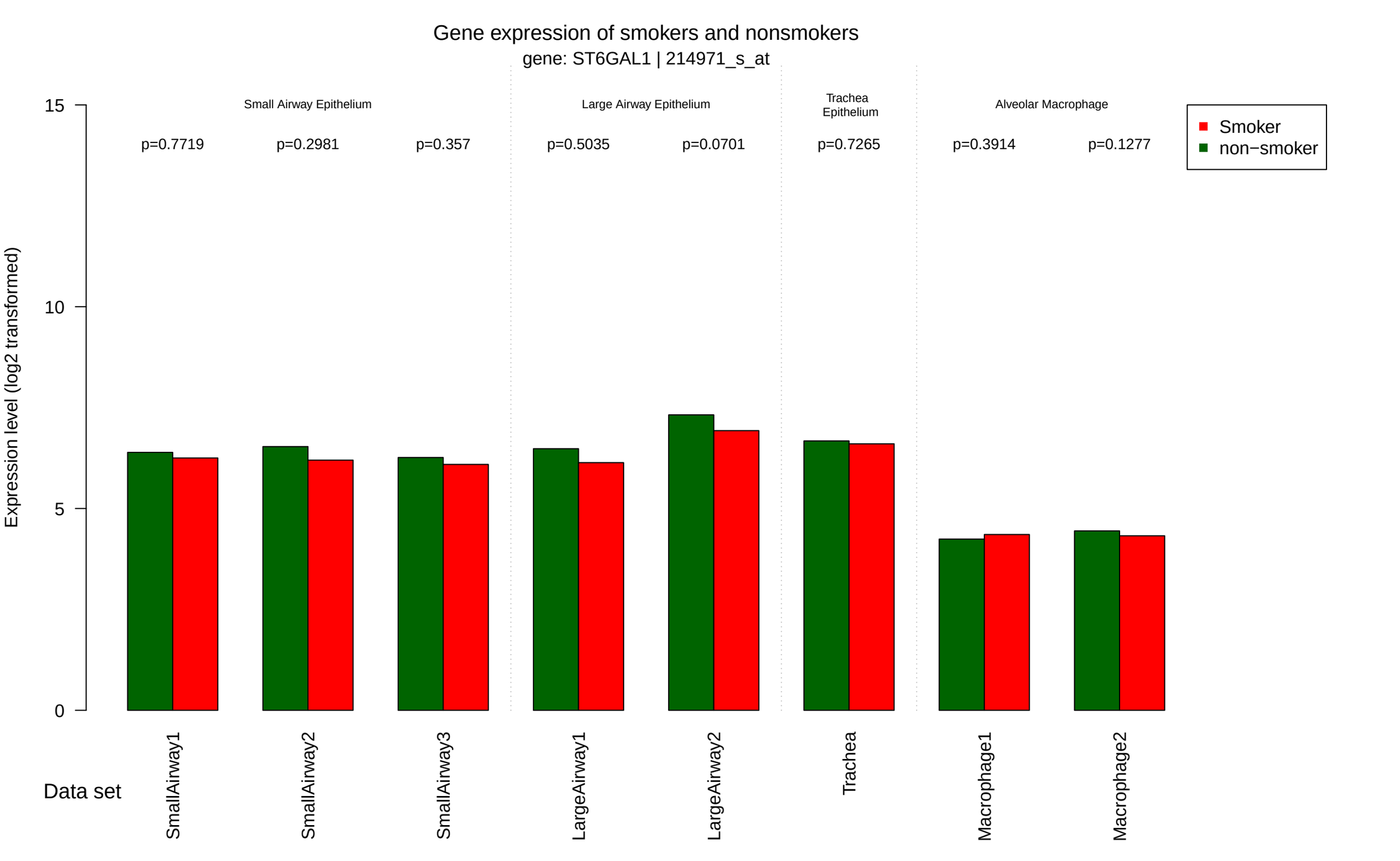
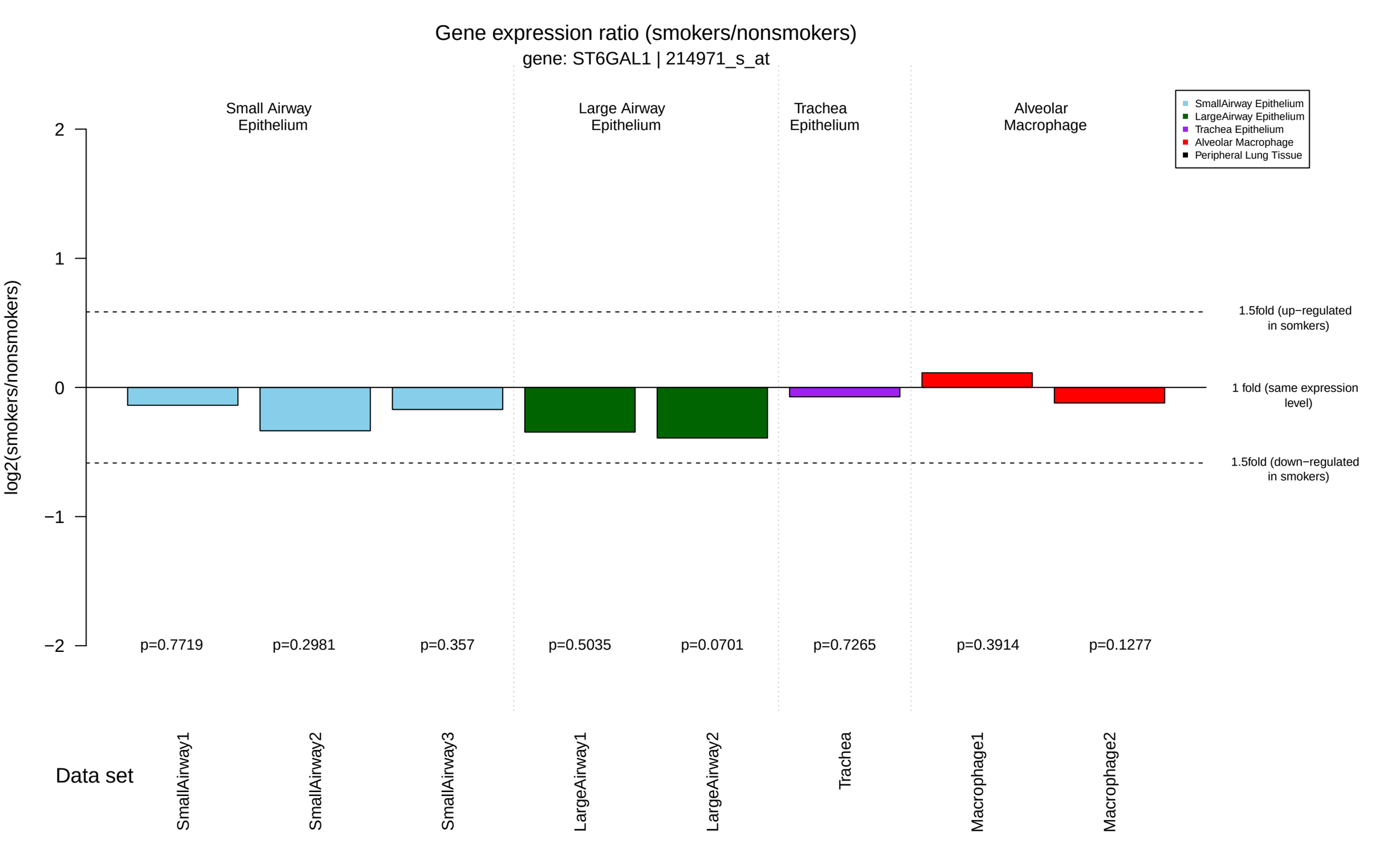
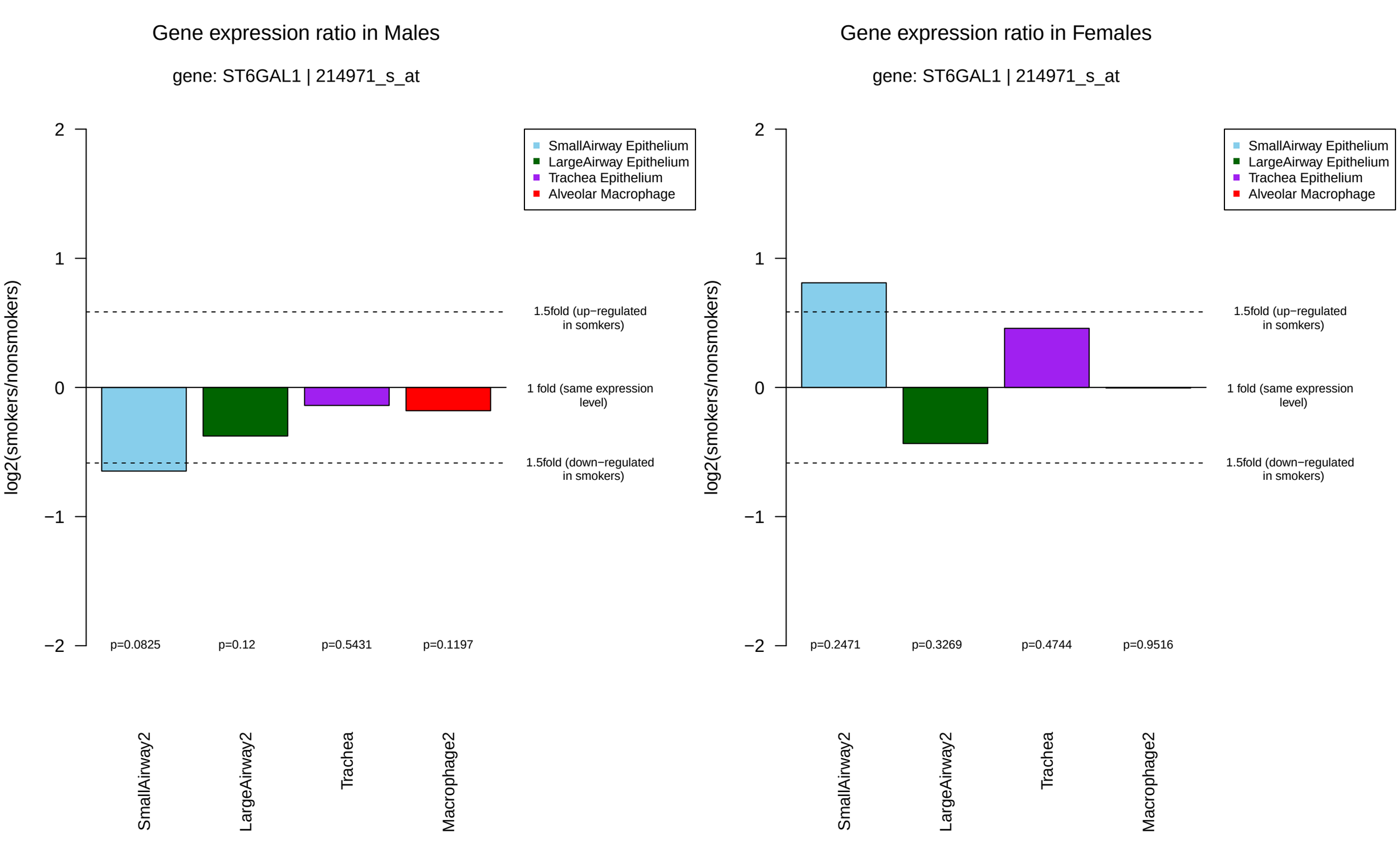
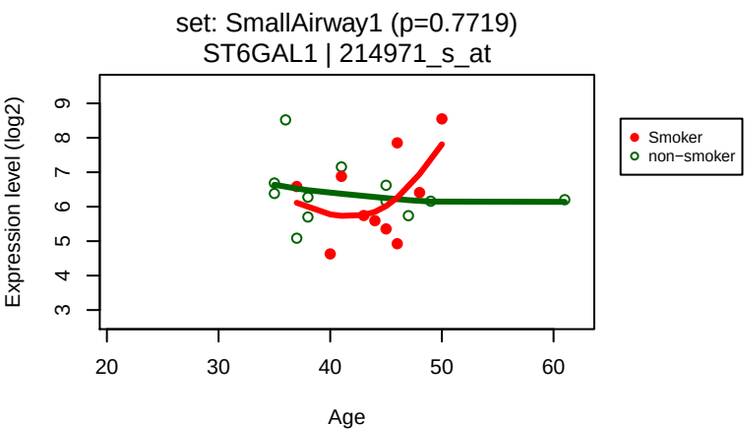
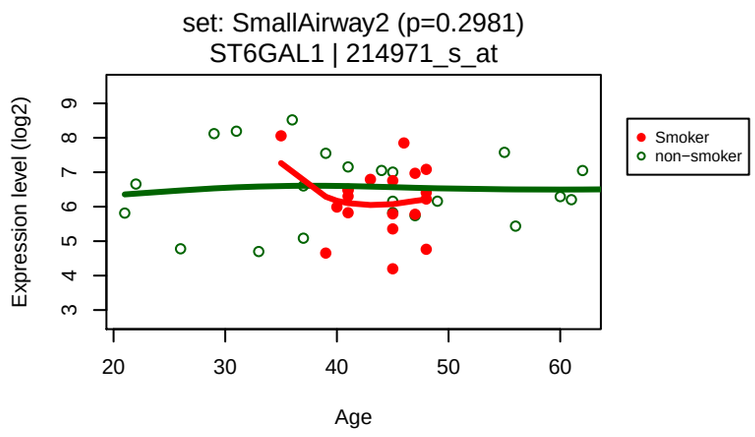
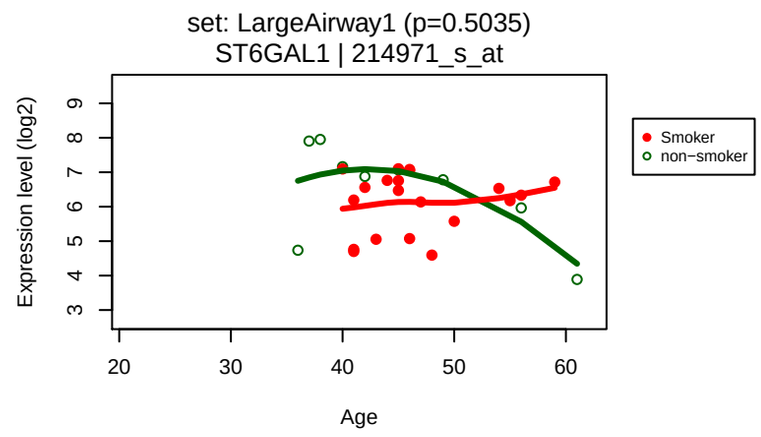

Gene expression of smokers and nonsmokers
gene: ST6GAL1 | 214971_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7719
p=0.2981
p=0.357
p=0.5035
p=0.0701
p=0.7265
p=0.3914
p=0.1277
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

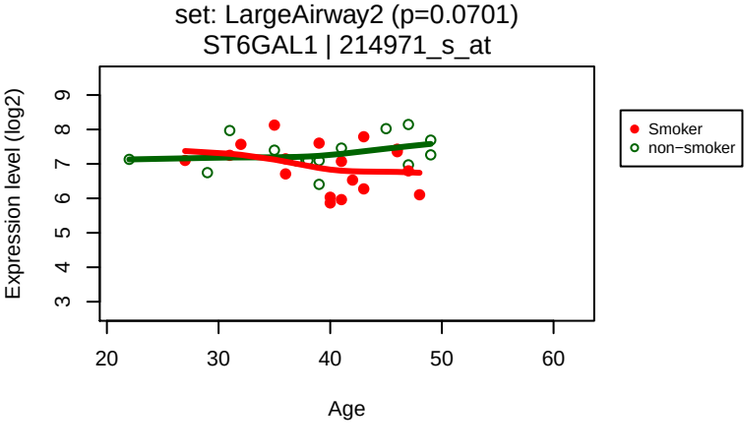

Gene expression ratio (smokers/nonsmokers)
gene: ST6GAL1 | 214971_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7719
p=0.2981
p=0.357
p=0.5035
p=0.0701
p=0.7265
p=0.3914
p=0.1277
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

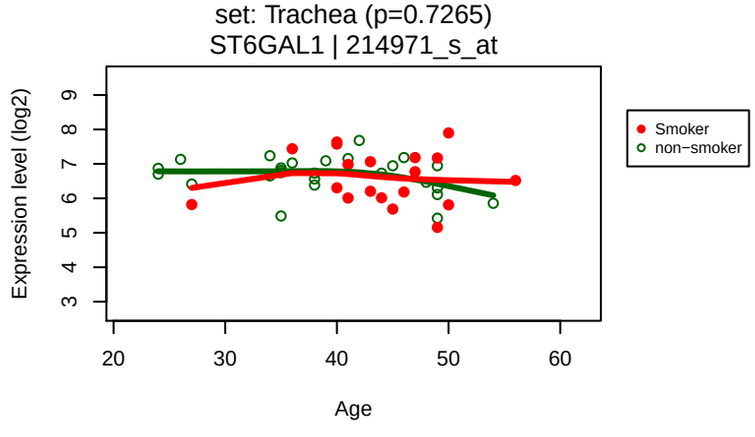

Gene expression ratio in Males
gene: ST6GAL1 | 214971_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0825
p=0.12
p=0.5431
p=0.1197
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GAL1 | 214971_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2471
p=0.3269
p=0.4744
p=0.9516
SmallAirway2
LargeAirway2
Trachea
Macrophage2

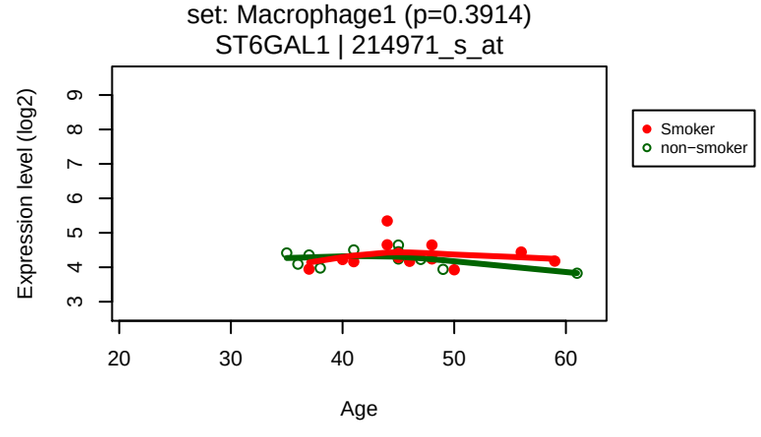

set: SmallAirway1 (p=0.7719)
ST6GAL1 | 214971_s_at
Expression level (log2)
Age
Smoker
non−smoker

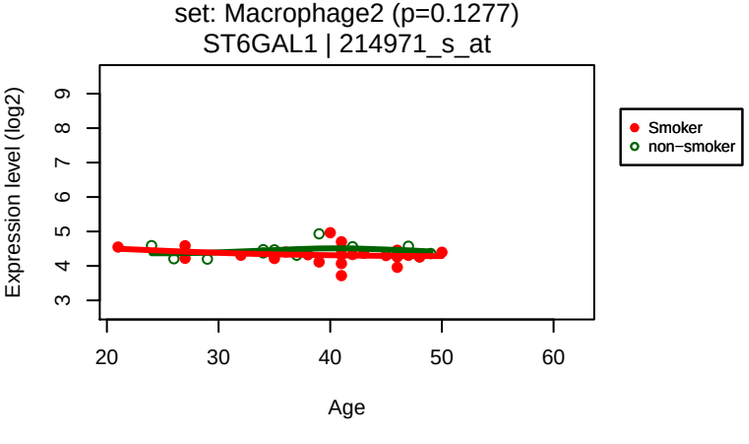

set: SmallAirway2 (p=0.2981)
ST6GAL1 | 214971_s_at
Expression level (log2)
Age
Smoker
non−smoker

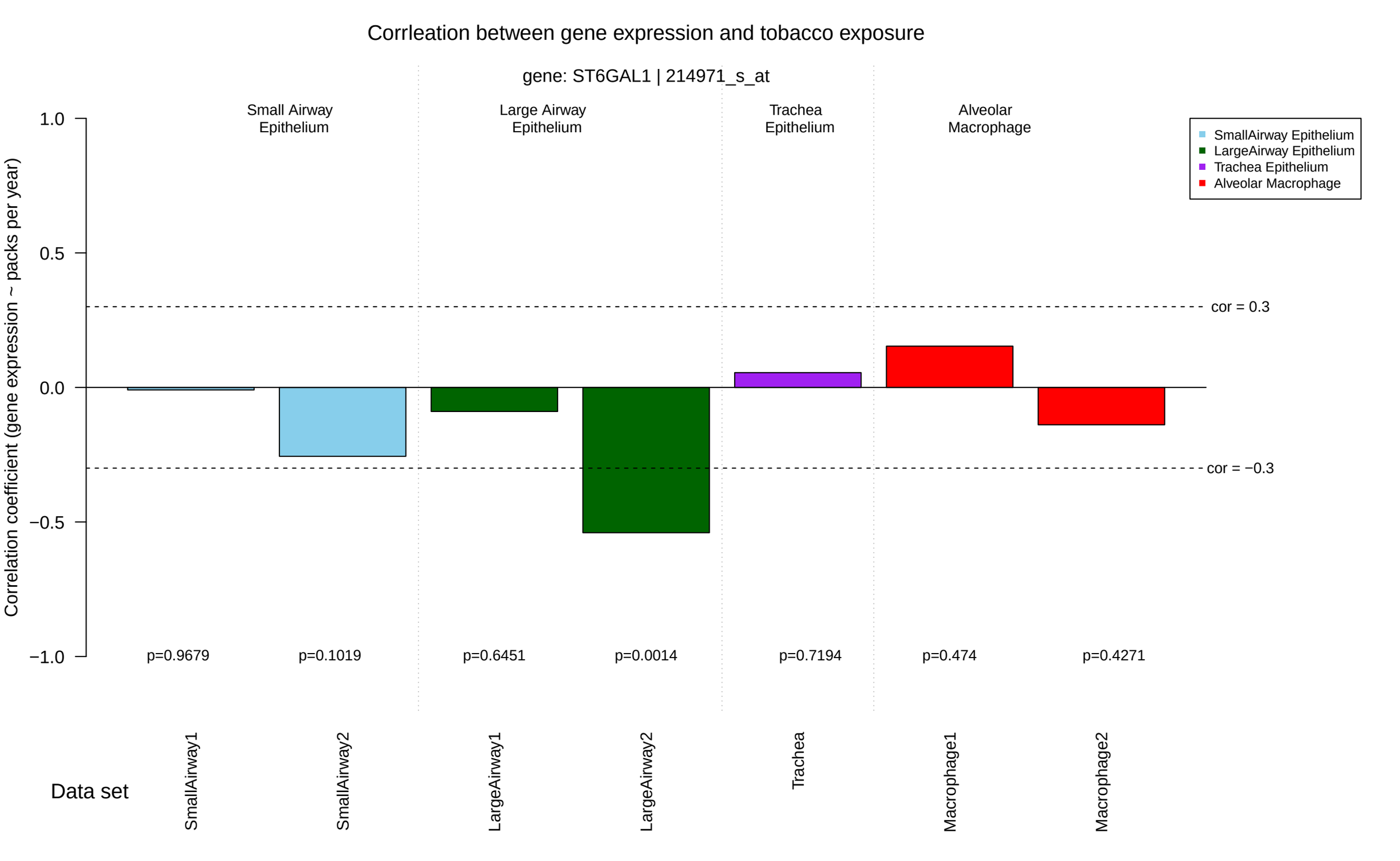

set: LargeAirway1 (p=0.5035)
ST6GAL1 | 214971_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0701)
ST6GAL1 | 214971_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7265)
ST6GAL1 | 214971_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3914)
ST6GAL1 | 214971_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1277)
ST6GAL1 | 214971_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GAL1 | 214971_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9679
p=0.1019
p=0.6451
p=0.0014
p=0.7194
p=0.474
p=0.4271
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2